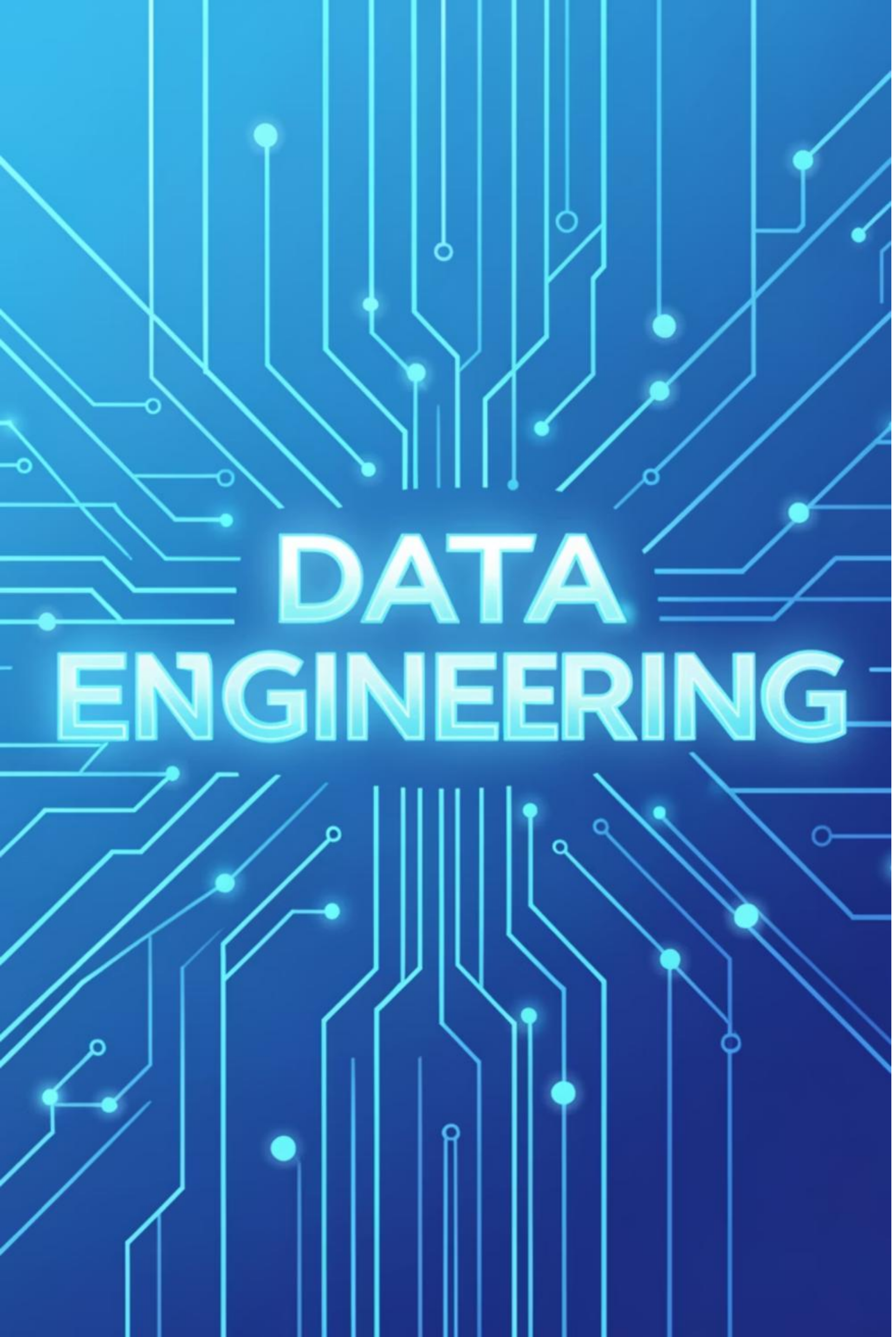

Data Engineering

Spring '26 – DS463

Feb 19, 2026

Lecture 15

Logistics: Your Course Essentials Essentials

Class Timings

Monday (LH6), Tuesday (LH8): 11:30 AM – 12:20 PM

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Credit Hours

2 hours (Lectures) + 1 hour (Practical (Practical Session)

Office Hours

Monday, Tuesday: 10:00 AM – 11:00 AM, AM, S18 2nd Floor NAB or by appointment.

Contact & Support

MS Teams: w5t93yk

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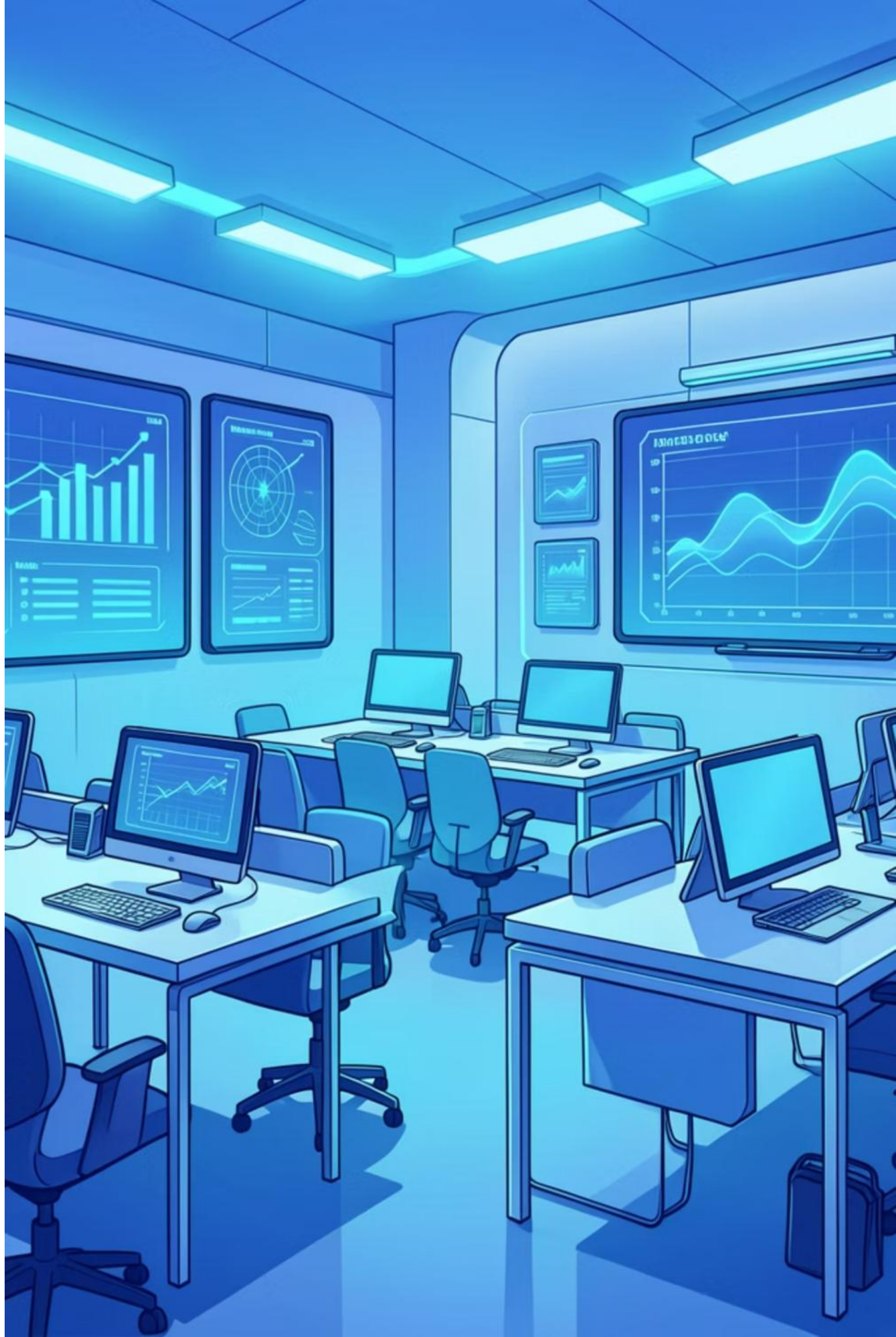


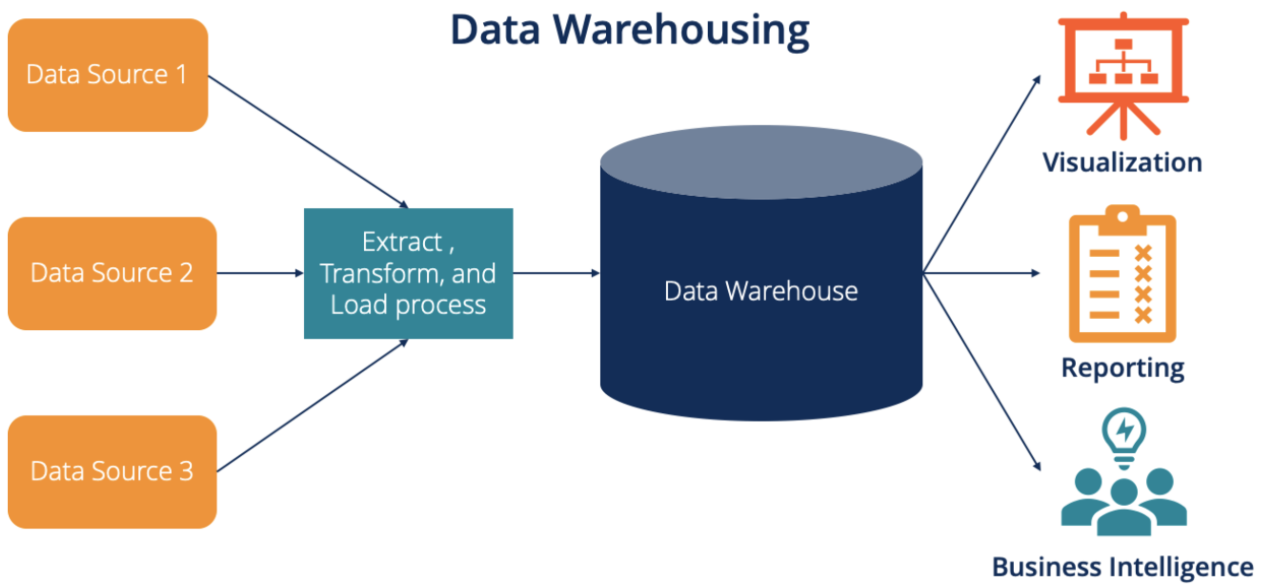
Major Architecture Concepts

- **Domain and Services**
- **Distributed Systems, Scalability and Design for failure**
- **Tight vs Loose Coupling**
- **User Access**
- **Event Driven Architecture**
- **Brownfield vs Greenfield Projects**

Examples and Types of Data Architecture

Data warehouse

- **A central data hub used for reporting and analytics.**
- **Data is highly structured and formatted for analysis.**
- First defined in 1989 by Bill Inmon as:
  - Subject-oriented
  - Integrated
  - Nonvolatile (not frequently changed)
  - Time-variant (tracks historical data)

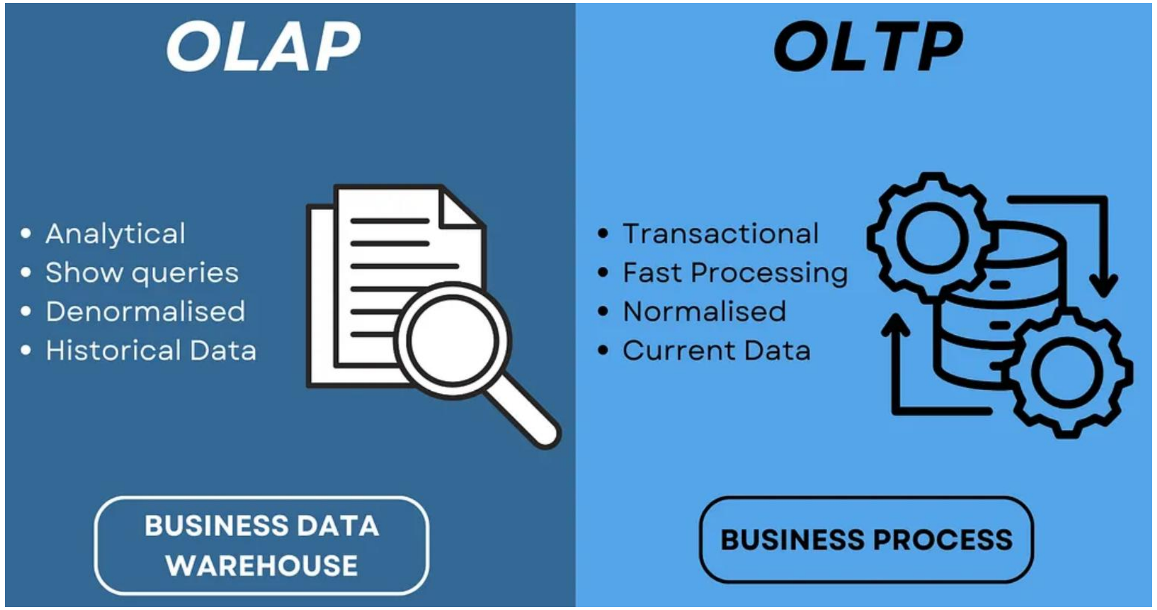


Examples and Types of Data Architecture

Data warehouse

Characteristics

- Separates OLAP from OLTP
- OLTP = transactional systems (production databases).
- OLAP = analytics systems
- Separating them:
    - Protects production systems.
    - Improves analytics performance.



https://blog.dataengineerthings.org/transactional-vs-analytical-a-deep-dive-into-oltp-and-olap-for-data-professionals-feb013a7264f

Examples and Types of Data Architecture

Data warehouse

MPP (Massively Parallel Processing)

Distributes queries across many machines.

Optimized for:

- Large data scans
- Aggregations
- Statistical calculations

Uses SQL (like relational databases).

Examples and Types of Data Architecture

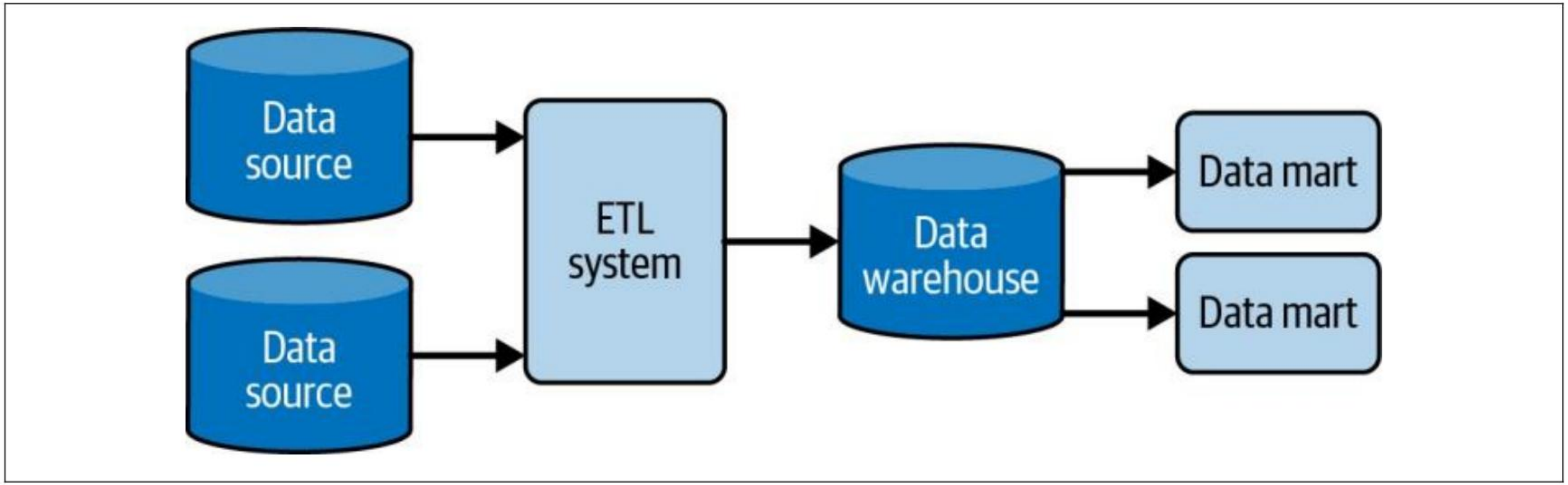


Figure 3-10. Basic data warehouse with ETL

Data Mart: Refined subset of data ware house.
Here further denormalized and transformed data specific to a subject area is present.
A data mart makes data more easily accessible to analysts and report developers.
Data marts provide an additional stage of transformation beyond that provided by the initial ETL or ELT pipelines.

Examples and Types of Data Architecture

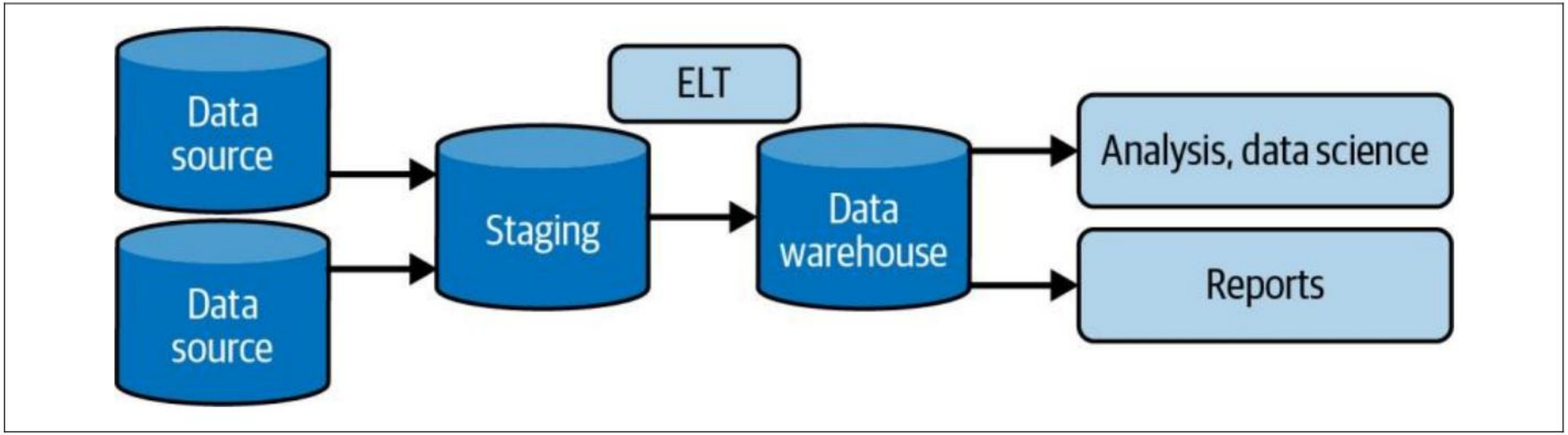


Figure 3-11. ELT—extract, load, and transform

Staging area: Contains raw data.

Examples and Types of Data Architecture

Evolution of Data warehouse

🏢 Traditional (On-Premises)

Used mainly by large enterprises.

Very expensive (millions of dollars).

Required large internal teams (DBAs, ETL developers).

Infrastructure was managed internally.

☁️ Modern (Cloud-Based)

Pay-as-you-go model.

Scalable and more affordable.

Infrastructure managed by cloud providers.

Smaller teams can manage complex data systems.

Examples and Types of Data Architecture

Feature	Data Warehouse	Data Lake	Data Lakehouse
Structured data	✅	✅	✅
Unstructured data	❌	✅	✅
ACID	✅	❌ (traditional)	✅
Schema-on-write	✅	❌	Hybrid
Schema-on-read	❌	✅	Hybrid
Storage cost	Higher	Low	Low
Centralized	Yes	Usually	Often

Examples and Types of Data Architecture

Lambda Architecture

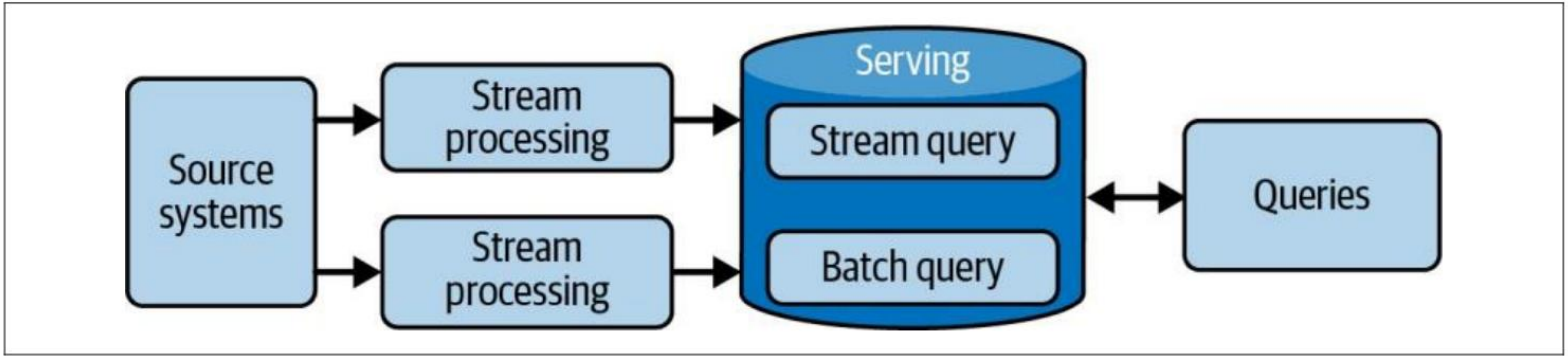


Figure 3-14. Lambda architecture

Examples and Types of Data Architecture

Lambda Architecture

Independent Layers: Lambda architecture separates processing into three independent systems**:**

- **Batch Layer** – handles large-scale, historical data processing.
- **Streaming Layer** – handles real-time, low-latency data.
- **Serving Layer** – provides combined query results from batch + streaming layers.

Data Source: Ideally immutable and append-only; data is sent to both batch and stream layers.

Main Idea: Combines speed (real-time) and accuracy (batch) in a single architecture.

Managing multiple systems with different codebases is as difficult as it sounds, creating error-prone systems with code and data that are extremely difficult to reconcile.

Examples and Types of Data Architecture

Kappa Architecture

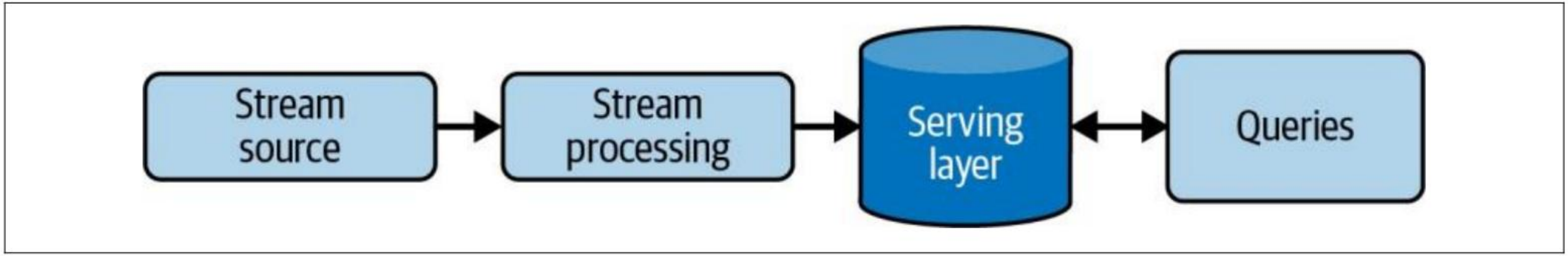
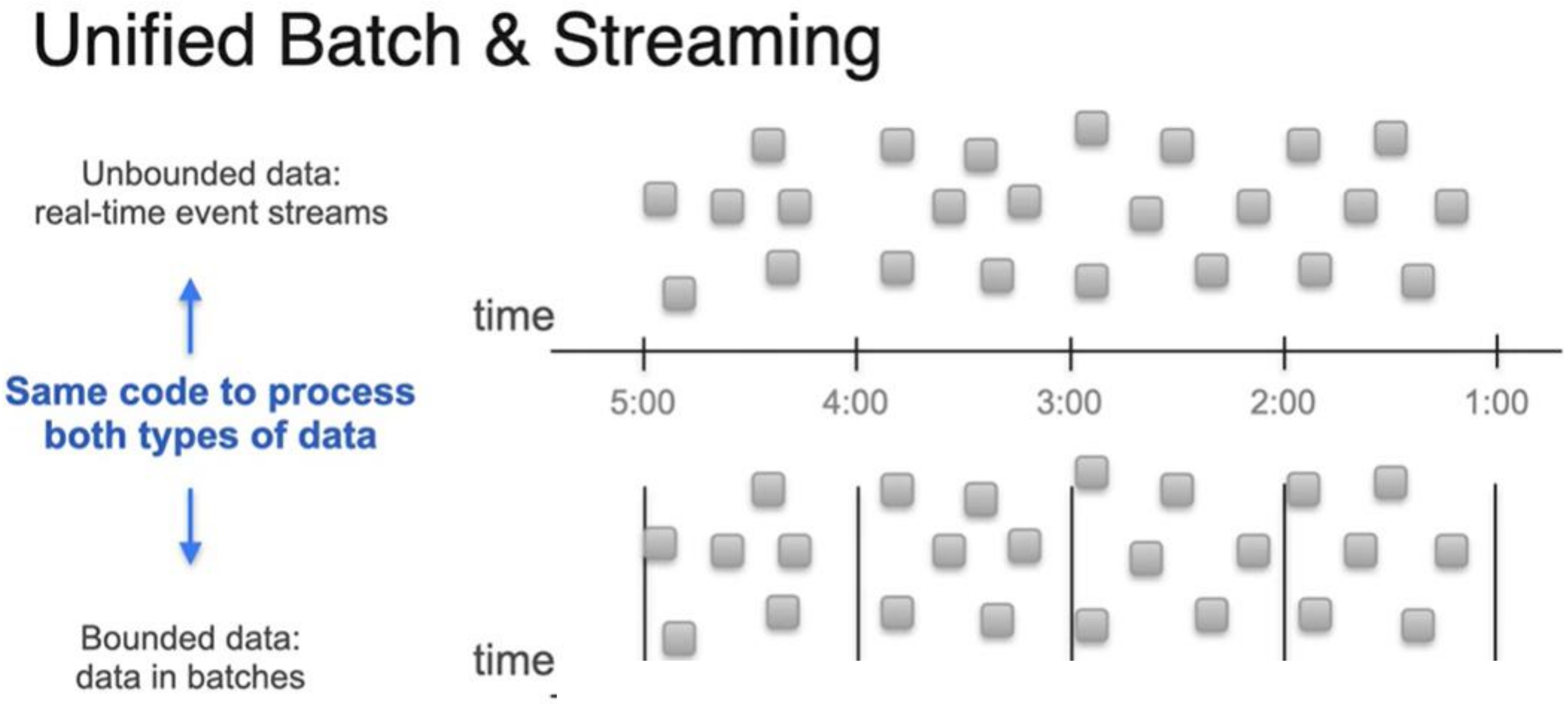


Figure 3-15. Kappa architecture

Real-time and batch processing can be applied seamlessly to the same data by reading the live event stream directly and replaying large chunks of data for batch processing.

Examples and Types of Data Architecture

The Dataflow Model and Unified Batch and Streaming



Examples and Types of Data Architecture

The Dataflow Model and Unified Batch and Streaming

Data as Events: Treats all data as a stream of events, whether real-time or batch

Bounded vs. Unbounded:

Unbounded streams → Real-time data with no fixed end.

Bounded streams → Batch data; boundaries create natural windows.

Windows for Aggregation:

Real-time aggregations are performed over windows.

Window types include:

Tumbling windows – fixed, non-overlapping intervals.

Sliding windows – overlapping intervals.

Unified Processing:

Both batch and real-time processing happen in the same system.

The same code can handle batch or streaming data.

Philosophy:"Batch is just a special case of streaming."